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Duration	7 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. The list must be sorted

c. There should be direct access to the middle element in any sublist

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The list is sorted

c. The search continues in the right sublist

d. The search ends successfully ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted

b. When the list is large and unsorted

c. When the list is small and sorted

d. When the list is small and unsorted ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search is more efficient than binary search on large lists.
d. Linear search requires the list to be sorted.

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the left sublist
c. In the right sublist ✓
d. In the middle of the list

Question **6**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the search algorithm finishes
c. When the list is sorted
d. When the element is found in the list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. Must use a sorted array
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. There must be a mechanism to access middle element directly

Question **8**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 500
c. 10 ✓
d. 100

Question **9**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Both (1) & (3)
c. Hash
d. Binary ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search backtracks to find duplicate elements
d. The search continues until the end of the list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Sorting the list before searching
c. Skipping every second element
d. Dividing the list in half repeatedly

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Binary search
c. Dynamic search
d. Linear search ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the left sublist ✓
c. At the beginning of the list
d. In the right sublist

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(log n)
c. O(1) ✓
d. O(n log n)

Question **15**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Bubble ✓

d. Complexity